WEEK 5 – HEALTHCARE DATA CLEANING & DATA MODELING

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1. Introduction

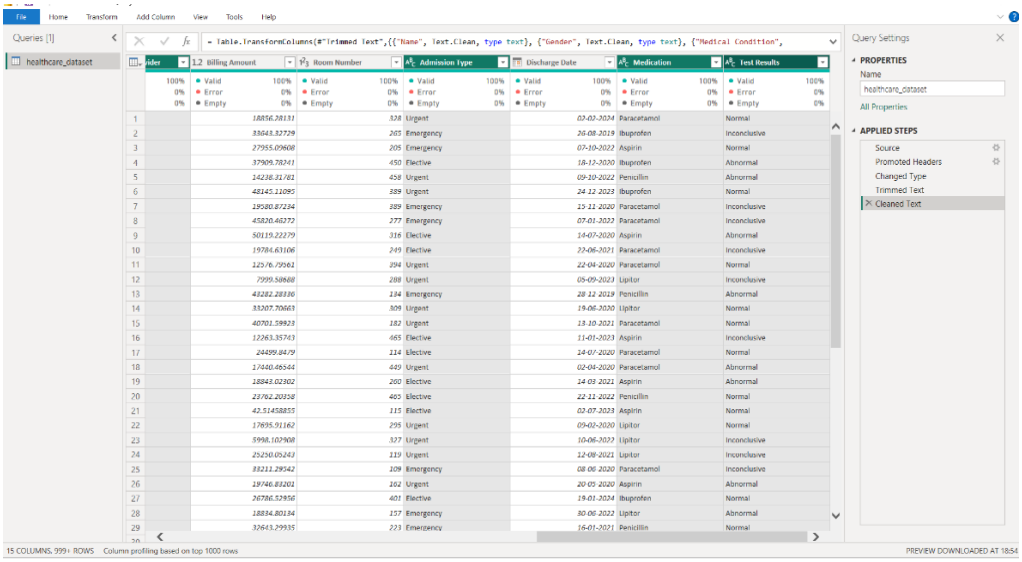
This project focuses on cleaning and modeling a Healthcare Dataset using Microsoft Power BI. The dataset was imported into Power Query, cleaned and analyzed. An Age Group column was created using conditional logic. Separate Dim_Patient, Dim_Admission and Billing tables were created from the cleaned dataset. The tables were then connected using Patient ID and Admission-ID to create a structured data model.

2. Objectives

- Import the Healthcare Dataset into Power BI.
- Clean and analyze the dataset using Power Query.
- Remove unnecessary spaces using Trim.
- Create an Age Group column.
- Create Dim_Patient and Dim_Admission tables.
- Create Patient ID and Admission-ID columns.
- Create a Billing table.
- Use Merge Queries to match related data.
- Create relationships between the tables.
- Build a structured healthcare data model.

3. Data Import and Cleaning

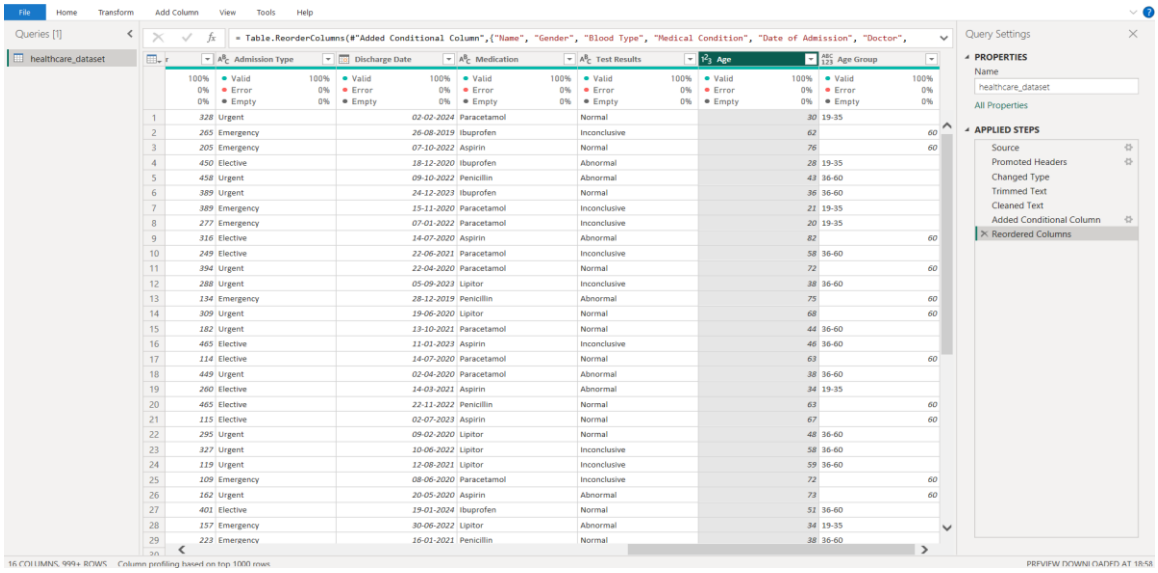
The Healthcare Dataset was imported into Power BI using Power Query Editor. The data was inspected for column names, data types, missing values and duplicate records. Text fields were cleaned using Trim and other transformation options to improve data consistency.



4. Creating Age Group

An Age Group column was created using conditional logic based on the patient's age. The patients were categorized into four groups: 0–18, 19–35, 36–60 and 60+.

Age Range	Age Group
0–18	0-18
19–35	19-35
36–60	36-60
61+	60+



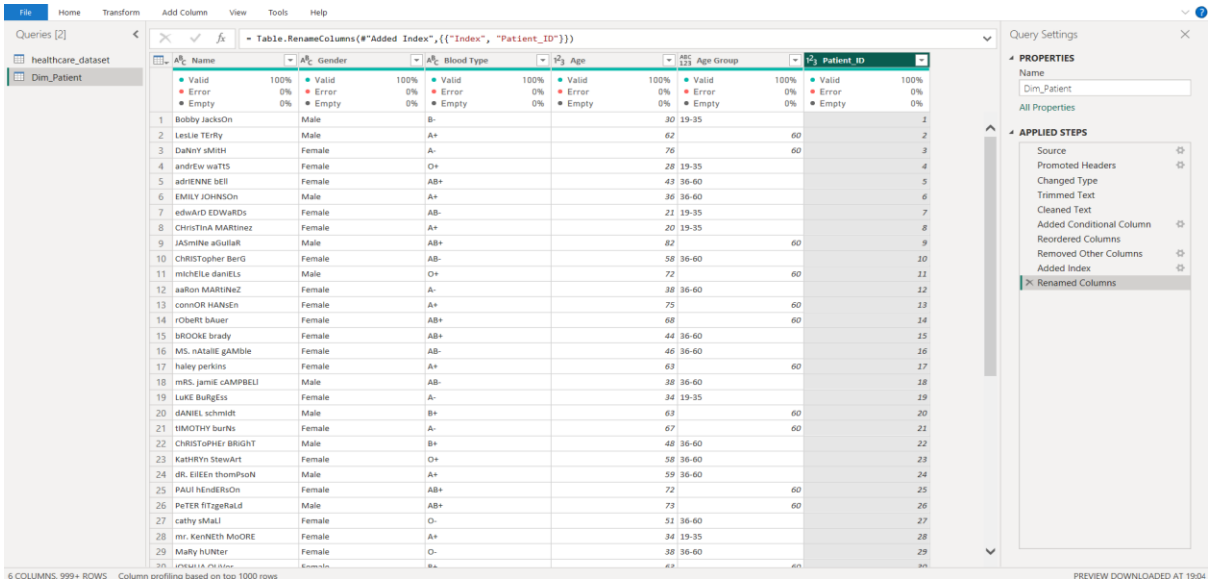
5. Creating Dim_Patient Table

The cleaned Healthcare Dataset was duplicated to create the Dim_Patient table. Patient-related columns such as Age, Age Group, Blood Type, Gender, Name and Patient ID were retained. Patient ID was created as the identifier used to connect patient information with the Billing table.

	Name	Gender	Blood Type	Age	Age Group
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	Bobby JacksOn	Male	B-	30	19-35
2	LesLie TErRy	Male	A+	62	60
3	DaNnY sMitH	Female	A-	76	60
4	andrEw waTtS	Female	O+	28	19-35
5	adrIENNE bEll	Female	AB+	43	36-60
6	EMILY JOHNSOn	Male	A+	36	36-60
7	edwArD EDWaRDs	Female	AB-	21	19-35
8	CHrisTInA MARtinez	Female	A+	20	19-35
9	JASmINe aGuIlaR	Male	AB+	82	60
10	ChRISTopher BerG	Female	AB-	58	36-60
11	mIchElLe daniELs	Male	O+	72	60
12	aaRon MARtiNeZ	Female	A-	38	36-60
13	connOR HANsEn	Female	A+	75	60
14	rObeRt bAuer	Female	AB+	68	60
15	bROOkE brady	Female	AB+	44	36-60
16	MS. nAtalIE gAMble	Female	AB-	46	36-60
17	haley perkins	Female	A+	63	60
18	mRS. jamiE cAMPBELl	Male	AB-	38	36-60
19	LuKE BuRgEss	Female	A-	34	19-35
20	dANIEL schmIdt	Male	B+	63	60
21	tIMOTHY burNs	Female	A-	67	60
22	ChRISToPHEr BRiGhT	Male	B+	48	36-60
23	KatHRYn StewArt	Female	O+	58	36-60
24	dR. EilEEn thomPsoN	Male	A+	59	36-60
25	PAUl hEndERsOn	Female	AB+	72	60
26	PeTER fiTzgeRaLd	Male	AB+	73	60
27	cathy sMaLl	Female	O-	51	36-60
28	mr. KenNEth MoORE	Female	A+	34	19-35
29	MaRy hUNter	Female	O-	38	36-60
30	jOSHUA OLiVer	Female	B+	63	60

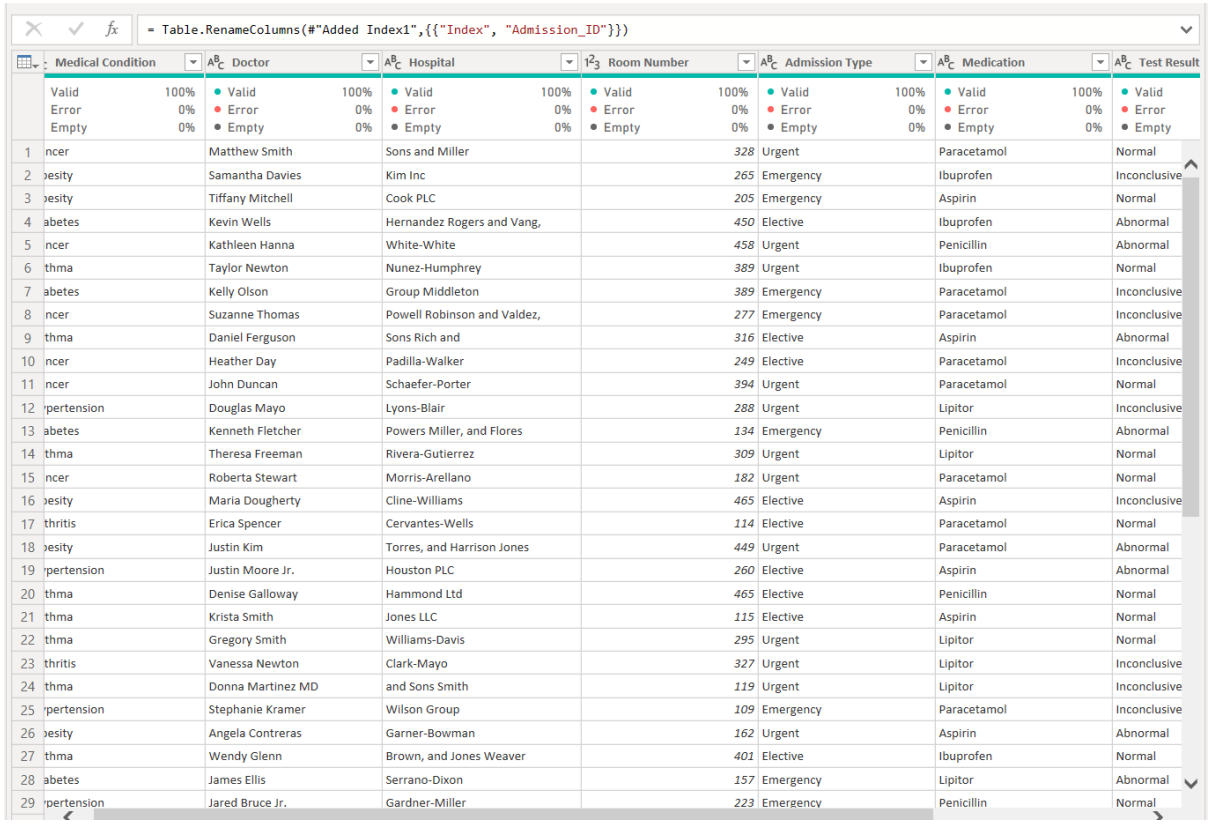
6. Creating Patient ID

A Patient ID column was created in the Dim_Patient table. This identifier was used to match patient information with the corresponding records in the Billing table.



7. Creating Dim_Admission Table

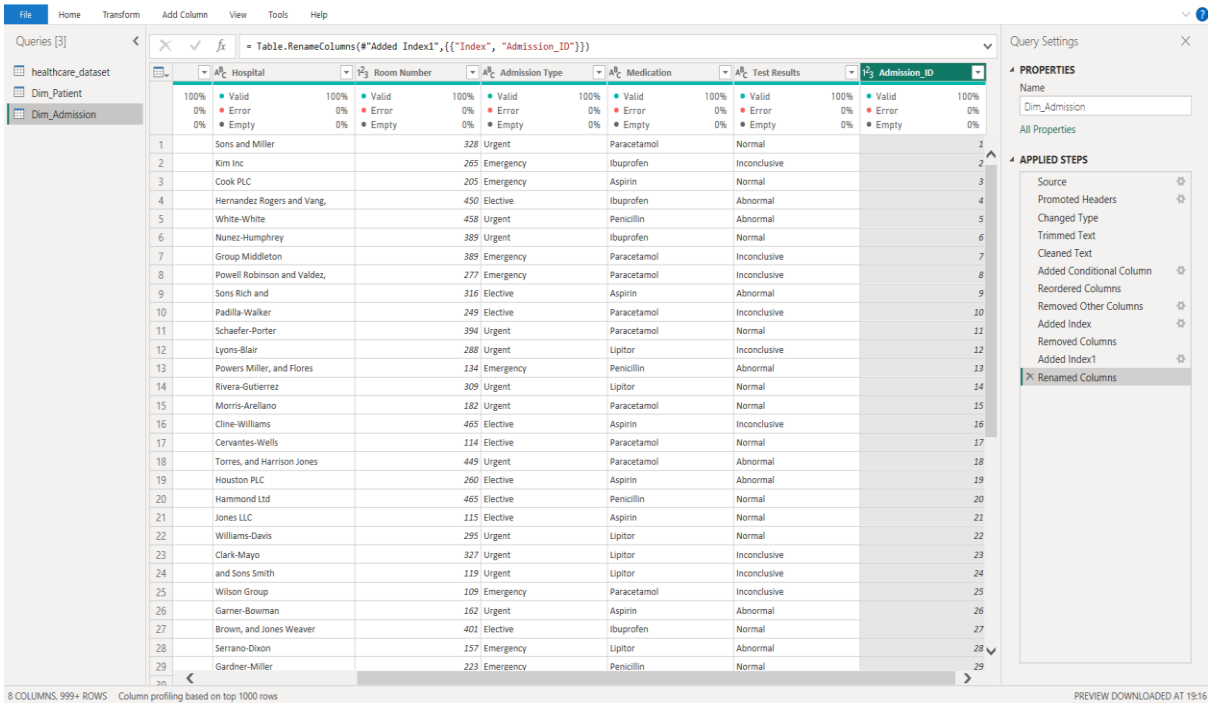
The cleaned Healthcare Dataset was duplicated to create the Dim_Admission table. Admission-related information such as Admission Type, Admission-ID, Doctor, Hospital, Medical Condition, Medication, Room Number and Test Results was retained.



	Medical Condition	Doctor	Hospital	Room Number	Admission Type	Medication	Test Result
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	ncer	Matthew Smith	Sons and Miller	328	Urgent	Paracetamol	Normal
2	esity	Samantha Davies	Kim Inc	265	Emergency	Ibuprofen	Inconclusive
3	esity	Tiffany Mitchell	Cook PLC	205	Emergency	Aspirin	Normal
4	abetes	Kevin Wells	Hernandez Rogers and Vang,	450	Elective	Ibuprofen	Abnormal
5	ncer	Kathleen Hanna	White-White	458	Urgent	Penicillin	Abnormal
6	thma	Taylor Newton	Nunez-Humphrey	389	Urgent	Ibuprofen	Normal
7	abetes	Kelly Olson	Group Middleton	389	Emergency	Paracetamol	Inconclusive
8	ncer	Suzanne Thomas	Powell Robinson and Valdez,	277	Emergency	Paracetamol	Inconclusive
9	thma	Daniel Ferguson	Sons Rich and	316	Elective	Aspirin	Abnormal
10	ncer	Heather Day	Padilla-Walker	249	Elective	Paracetamol	Inconclusive
11	ncer	John Duncan	Schaefer-Porter	394	Urgent	Paracetamol	Normal
12	rpertension	Douglas Mayo	Lyons-Blair	288	Urgent	Lipitor	Inconclusive
13	abetes	Kenneth Fletcher	Powers Miller, and Flores	134	Emergency	Penicillin	Abnormal
14	thma	Theresa Freeman	Rivera-Gutierrez	309	Urgent	Lipitor	Normal
15	ncer	Roberta Stewart	Morris-Arellano	182	Urgent	Paracetamol	Normal
16	esity	Maria Dougherty	Cline-Williams	465	Elective	Aspirin	Inconclusive
17	thritis	Erica Spencer	Cervantes-Wells	114	Elective	Paracetamol	Normal
18	esity	Justin Kim	Torres, and Harrison Jones	449	Urgent	Paracetamol	Abnormal
19	rpertension	Justin Moore Jr.	Houston PLC	260	Elective	Aspirin	Abnormal
20	thma	Denise Galloway	Hammond Ltd	465	Elective	Penicillin	Normal
21	thma	Krista Smith	Jones LLC	115	Elective	Aspirin	Normal
22	thma	Gregory Smith	Williams-Davis	295	Urgent	Lipitor	Normal
23	thritis	Vanessa Newton	Clark-Mayo	327	Urgent	Lipitor	Inconclusive
24	thma	Donna Martinez MD	and Sons Smith	119	Urgent	Lipitor	Inconclusive
25	rpertension	Stephanie Kramer	Wilson Group	109	Emergency	Paracetamol	Inconclusive
26	esity	Angela Contreras	Garner-Bowman	162	Urgent	Aspirin	Abnormal
27	thma	Wendy Glenn	Brown, and Jones Weaver	401	Elective	Ibuprofen	Normal
28	abetes	James Ellis	Serrano-Dixon	157	Emergency	Lipitor	Abnormal
29	rpertension	Jared Bruce Jr.	Gardner-Miller	223	Emergency	Penicillin	Normal

8. Creating Admission-ID

An Admission-ID column was created in the Dim_Admission table. It was used as the identifier for connecting admission information with the Billing table.



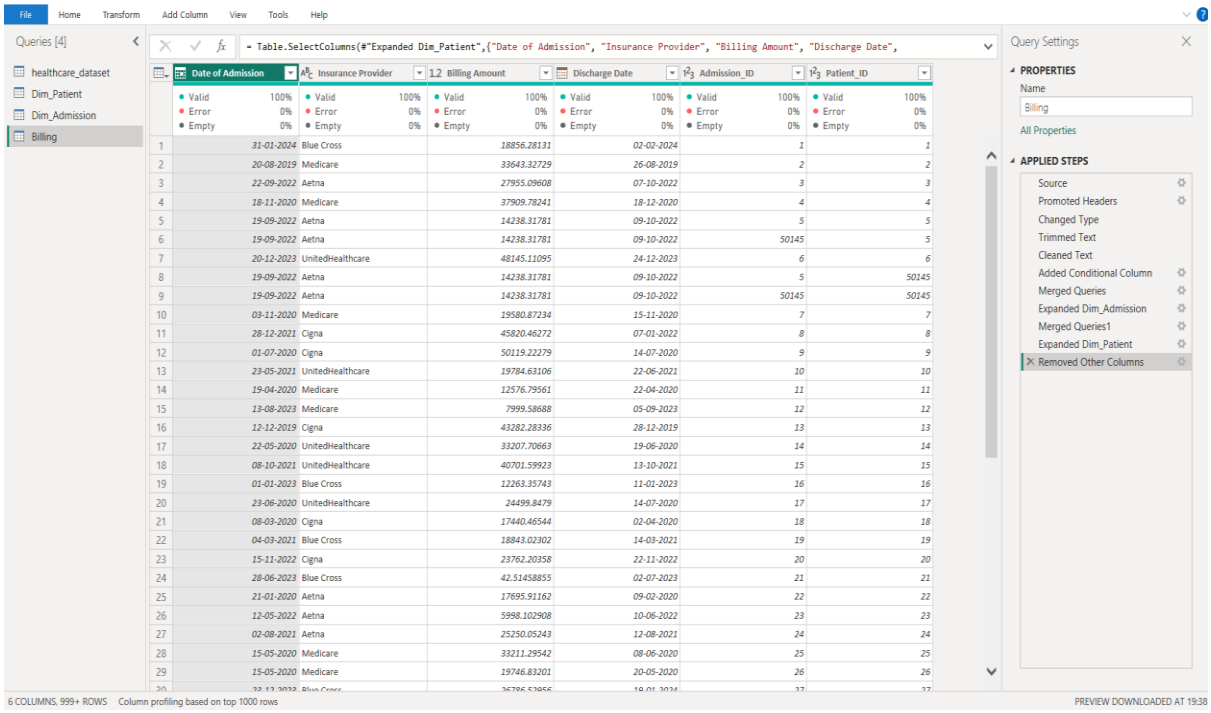
9. Creating Billing Table

A Billing table was created from the cleaned Healthcare Dataset. It contains Admission-ID, Billing Amount, Date of Admission, Discharge Date, Insurance Provider and Patient ID. The Billing table acts as the central table for analyzing healthcare billing information.

	Date of Admission	Insurance Provider	Billing Amount	Discharge Date
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	31-01-2024	Blue Cross	18856.28131	02-02-2024
2	20-08-2019	Medicare	33643.32729	26-08-2019
3	22-09-2022	Aetna	27955.09608	07-10-2022
4	18-11-2020	Medicare	37909.78241	18-12-2020
5	19-09-2022	Aetna	14238.31781	09-10-2022
6	19-09-2022	Aetna	14238.31781	09-10-2022
7	20-12-2023	UnitedHealthcare	48145.11095	24-12-2023
8	19-09-2022	Aetna	14238.31781	09-10-2022
9	19-09-2022	Aetna	14238.31781	09-10-2022
10	03-11-2020	Medicare	19580.87234	15-11-2020
11	28-12-2021	Cigna	45820.46272	07-01-2022
12	01-07-2020	Cigna	50119.22279	14-07-2020
13	23-05-2021	UnitedHealthcare	19784.63106	22-06-2021
14	19-04-2020	Medicare	12576.79561	22-04-2020
15	13-08-2023	Medicare	7999.58688	05-09-2023
16	12-12-2019	Cigna	43282.28336	28-12-2019
17	22-05-2020	UnitedHealthcare	33207.70663	19-06-2020
18	08-10-2021	UnitedHealthcare	40701.59923	13-10-2021
19	01-01-2023	Blue Cross	12263.35743	11-01-2023
20	23-06-2020	UnitedHealthcare	24499.8479	14-07-2020
21	08-03-2020	Cigna	17440.46544	02-04-2020
22	04-03-2021	Blue Cross	18843.02302	14-03-2021
23	15-11-2022	Cigna	23762.20358	22-11-2022
24	28-06-2023	Blue Cross	42.51458855	02-07-2023
25	21-01-2020	Aetna	17695.91162	09-02-2020
26	12-05-2022	Aetna	5998.102908	10-06-2022
27	02-08-2021	Aetna	25250.05243	12-08-2021
28	15-05-2020	Medicare	33211.29542	08-06-2020
29	15-05-2020	Medicare	19746.83201	20-05-2020
30	22-12-2023	Blue Cross	26786.52956	18-01-2024

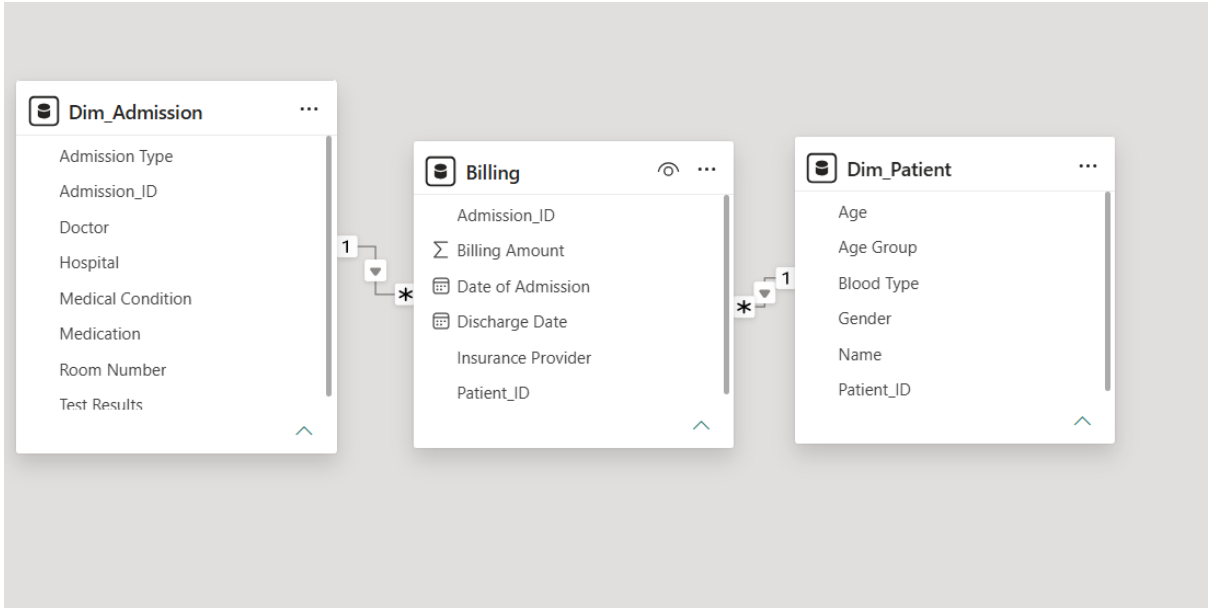
10. Using Merge Queries

Merge Queries was used to match related records using common identifier columns. Patient ID was used to match patient information, while Admission-ID was used to match admission information. This helped prepare the tables for the final Power BI data model.



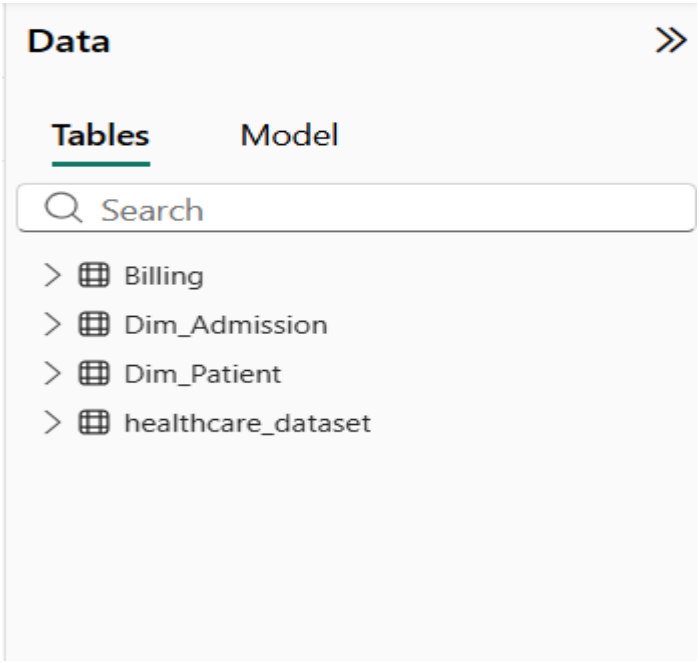
11. Creating Relationships

After preparing the tables, relationships were created in Power BI Model View. Dim_Patient was connected to Billing using Patient ID, and Dim_Admission was connected to Billing using Admission-ID. Both relationships were configured as one-to-many relationships.



12. Final Data Model

The final data model contains three tables: Dim_Patient, Dim_Admission and Billing. Dim_Patient provides patient information, Dim_Admission provides admission information, and Billing contains billing-related information. The dimension tables are connected to the Billing table through Patient ID and Admission-ID using one-to-many relationships.



14. Conclusion

In Week 5, I successfully performed healthcare data cleaning and data modeling using Power BI. The dataset was imported and cleaned using Power Query. Trim and other transformations were used to improve data quality. An Age Group column was created to classify patients into different age categories. Dim_Patient, Dim_Admission and Billing tables were created by duplicating the cleaned dataset. Patient ID and Admission-ID were used as connecting keys. Finally, one-to-many relationships were created between the dimension tables and Billing table. The completed data model is ready for further healthcare analysis and dashboard development.